



Scientific collaborator recommendation via hypergraph embedding

Xiaochen Wang^a, Wensheng Huang^b, Butian Zhao^c, Shijuan Li^a,*

^a Department of Information Management, Peking University, Beijing, 100871, China

^b Department of Information Management, Beijing Jiaotong University, Beijing, 100044, China

^c Department of Big Data Management and Application, Beijing University of Chinese Medicine, Beijing, 100029, China

ARTICLE INFO

Dataset link: <http://doi.org/10.1145/1401890.1402008>

Keywords:

Collaborator recommendation
Hypergraph embedding
Translational distance models
Recommendation interpretability

ABSTRACT

Identifying potential scientific collaborators is critical to fostering innovation in an era of academic digitalization. Existing recommendation methods often rely on pairwise relations and fail to model the high-order, multi-relational nature of real-world collaboration networks. To address this, we propose a hypergraph embedding-based framework that constructs a heterogeneous Scientific Collaboration Hypergraph from the AMiner dataset. Using a hypergraph neural network and translational scoring, our method captures structural semantics and interdisciplinary patterns. The resulting graph contains 6,119 scholars, 18,092 publications, and nine types of hyperedges modeling diverse academic relations. Experimental results show that our approach achieves a Recall@10 of 0.1802, representing a 78% improvement over the strongest baseline. It also performs robustly in cold-start scenarios and generalizes well to interdisciplinary recommendations. A user study confirms the interpretability of the system, with *Usefulness* and *Trust* receiving average scores above 4.0 on a 5-point Likert scale. The proposed method demonstrates both effectiveness and transparency in collaborator recommendation.

1. Introduction

Scientific collaboration plays a central role in advancing knowledge and fostering innovation, particularly as research becomes increasingly interdisciplinary and data-intensive (Hara et al., 2003; Tortoriello et al., 2015). Despite the growing availability of academic social platforms, most researchers continue to rely on familiar coauthorship networks to initiate new collaborations (Lerner & Hâncean, 2023). This reliance restricts access to broader scholarly communities and highlights the need for intelligent, data-driven collaborator recommendation systems.

Conventional approaches typically adopt content-based filtering, which leverages textual features such as research topics and publication records (He et al., 2022; Tang et al., 2008), or collaborative filtering, which relies on historical coauthorship behavior (Konstas et al., 2009; Wang et al., 2022). While increasingly enhanced by deep learning techniques, these methods often struggle to capture the high-order and multi-relational structure of academic collaboration networks (Zhang et al., 2024). Moreover, they tend to underperform in cold-start scenarios where historical data is sparse, and they often lack the ability to generalize across disciplinary boundaries. These two challenges are particularly important in practical applications such as supporting early-career researchers and facilitating interdisciplinary team formation (Liu et al., 2024).

Graph-based models provide a structured framework for recommendation and have gained popularity for academic network representation (Yu et al., 2025). However, traditional graphs encode only pairwise relations and are inadequate for modeling collaborative events involving multiple entities. For example, a paper coauthored by Kathy, Joey, and Bob would be decomposed

* Correspondence to: Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing, China.

E-mail addresses: xcwang25@stu.pku.edu.cn (X. Wang), 23111187@bjtu.edu.cn (W. Huang), btzhao@bucm.edu.cn (B. Zhao), shijuan.li@pku.edu.cn (S. Li).

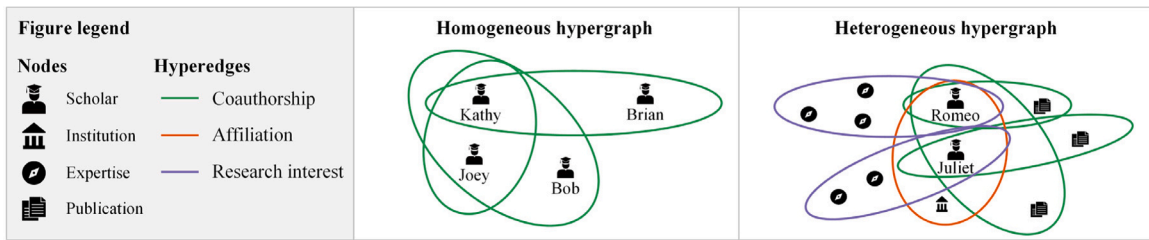


Fig. 1. An illustration of homogeneous and heterogeneous hypergraphs. The homogeneous hypergraph includes only coauthorship as the hyperedge type, while the heterogeneous hypergraph incorporates additional hyperedge types, such as institutional affiliations and research interests.

into three dyadic links (Kathy–Joey, Kathy–Bob, Joey–Bob), fragmenting the original group structure and obscuring the semantics of collective collaboration. This representation fails to reflect whether these scholars coauthored a single paper together or merely share pairwise ties, thereby limiting the model’s ability to capture high-order collaboration dynamics, particularly in multi-author, multi-institution, or cross-disciplinary contexts.

Hypergraphs naturally address this limitation by allowing each hyperedge to connect more than two nodes, thereby preserving the complete set of participants and the integrity of the collaborative event. In our setting, each hyperedge directly encodes a group-based relation such as coauthorship, meaning that the connection among Kathy, Joey, and Bob is modeled as a single semantic unit rather than fragmented links. This enables the model to learn interaction patterns at the event level. Furthermore, heterogeneous hypergraphs extend this capability by allowing different types of hyperedges to coexist within the same structure. These typed hyperedges enhance the semantic expressiveness of the graph and provide a unified representation of diverse academic contexts (Zhang et al., 2023). Fig. 1 illustrates the distinction between homogeneous and heterogeneous hypergraph modeling.

Compared with deep learning-based content models and collaborative filtering approaches, which often operate as black boxes (Silva Rodrigues et al., 2025), hypergraph-based methods offer both structural expressiveness and interpretability. Each hyperedge encodes a meaningful real-world collaboration context, enabling the generation of recommendation rationales grounded in observable academic relations. Recent pre-trained language models (PLMs), such as BERT and GPT variants, represent a distinct modeling paradigm. Instead of explicitly encoding structural collaboration graphs, they primarily rely on unstructured textual features to learn semantic similarity in vector space (Lin et al., 2025). While autoregressive PLMs like GPT can generate textual rationales when prompted, their underlying representations are still learned from flat sequences rather than structural graphs, making it difficult to model high-order multi-entity collaboration events in a principled and consistent manner (Wang et al., 2023). In summary, hypergraphs provide a principled and transparent modeling framework for collaborator recommendation, especially under data sparsity or interdisciplinary complexity.

To address these challenges, including inadequate structural modeling, limited robustness in cold-start scenarios, and insufficient adaptability to interdisciplinary contexts, this study is guided by the following research **questions**:

- How can a heterogeneous hypergraph be constructed to capture the high-order, multi-relational nature of scientific collaboration?
- Can hypergraph embeddings effectively identify suitable collaborators, particularly under cold-start and interdisciplinary conditions?
- How can interpretability be integrated into the recommendation pipeline to enhance transparency and user trust?

In response to these questions, we construct a novel Scientific Collaboration Hypergraph (SCH) based on the AMiner dataset and design a hypergraph embedding-based recommendation framework. A hypergraph neural network is employed to learn structure-aware scholar embeddings, and a translational scoring mechanism is used to infer potential collaborators. To enhance interpretability, we further generate natural language explanations grounded in predefined hyperedge relation templates. Experimental results demonstrate that our method outperforms baselines in both recommendation accuracy and user-perceived interpretability, and shows particularly strong performance in cold-start scenarios and interdisciplinary collaboration settings. This work makes the following **contributions** beyond existing methods:

- We define a heterogeneous Scientific Collaboration Hypergraph that preserves collaboration events as single hyperedges and integrates institutions, venues, and expertise, avoiding information loss from pairwise decomposition.
- We design a spectral hypergraph embedding framework with type-aware attention and a translational scoring mechanism tailored to multi-entity collaboration structures, enabling scalable and structure-aware ranking.
- We introduce an explanation module that converts learned importance over hyperedge types into natural language rationales, improving transparency and user trust without sacrificing accuracy.
- Experiments on AMiner show consistent gains over strong baselines in Recall@k and NDCG@k, with particularly strong results for cold-start scholars and interdisciplinary recommendations.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the proposed methodology. Sections 4 and 5 present the experimental setup and results. Section 6 concludes with implications and directions for future research. The code to reproduce the experiments is available at [GitHub](#).

2. Literature review

Scientific collaborator recommendation lies at the intersection of academic recommender systems and collaboration analysis. This section presents a structured review from three perspectives. First, we summarize recent advances in academic recommendation methods. Second, we review key factors influencing scientific collaboration, which inform the design of our model. Finally, we outline a taxonomy of hypergraph-based recommendation techniques and assess their applicability.

2.1. Recent advances in academic recommendation

Existing academic recommendation methods can be broadly grouped into three categories: traditional collaborative and content-based approaches, graph-based recommendation models, and pretrained language model-based methods.

Collaborative filtering methods model historical interactions, such as coauthorship or citation networks, to predict future links. For instance, Neural Collaborative Filtering (NCF) captures nonlinear user-item interactions through a multi-layer perceptron architecture, outperforming shallow matrix factorization techniques (He et al., 2017). Variational Autoencoders (Mult-VAE) extend this by modeling user preferences as probabilistic latent variables, enabling better generalization in sparse recommendation scenarios (Liang et al., 2018).

Graph-based methods further enrich the representation by exploiting the structural properties of academic networks. Neural Graph Collaborative Filtering (NGCF) incorporates higher-order neighborhood information through recursive message passing (Wang et al., 2019), while LightGCN simplifies the model by removing nonlinearities and feature transformations, focusing on weighted aggregation (He et al., 2020). Among these, Graph Filtering for Collaborative Filtering (GF-CF) (Shen et al., 2021) demonstrates state-of-the-art performance in OAG-Bench by integrating spectral graph filtering into GCN-like models to enhance the smoothness and discriminability of learned embeddings (Zhang et al., 2024). Although heterogeneous graphs can encode multiple relation types, they typically factorize a collaboration event into star shaped links or metapaths, which may weaken event level signals.

Pretrained models have also been applied to academic recommendation, with OAG-BERT being a representative example. It combines transformer-based semantic encoding of paper abstracts and titles with structural context from academic knowledge graphs, enabling zero-shot and few-shot recommendation capabilities (Liu et al., 2022). While powerful, OAG-BERT encodes coauthorship relations as independent triples during training, without explicitly modeling the coauthorship event as a unified, multi-entity structure. As a result, it may overlook the high-order and group-level semantics essential for capturing interdisciplinary and team-based scientific collaborations.

Despite notable progress, existing academic recommendation methods are approaching a performance ceiling. Overcoming this bottleneck requires incorporating richer academic information through more expressive graph structures (Wang et al., 2024; Zhang et al., 2024). To design such a structure effectively, it is first necessary to understand the underlying factors that influence scientific collaboration, which we review in the next section.

2.2. Scientific collaboration influencing factors

The factors influencing scientific collaboration have been the focus of research for decades, driven by the continuous evolution of collaborative practices in response to disciplinary, institutional, and technological changes (Hou et al., 2021).

Individual scholar attributes, such as productivity, research interests, career stage, and institutional affiliation, play a critical role in shaping collaboration dynamics. Highly productive researchers often occupy central positions within collaboration networks, making them more likely to attract new collaborators (Ebadi & Schiffauerova, 2015). Gender differences also influence collaboration patterns (Hill & Hincapie, 2025), while early-career researchers often act as bridges connecting disparate communities (Liu et al., 2024; Zou et al., 2023). Homophily, the tendency of scholars to collaborate with individuals who share similar attributes, significantly increases the likelihood of collaboration (Zhang et al., 2018). Moreover, alignment in research domains and the complementarity of academic differences have been identified as key drivers in establishing initial collaborations (Peng & Shi, 2024).

Beyond individual attributes, institutional and geographic factors further shape collaboration patterns. Institutional resources, such as funding and faculty networks, play a critical role in facilitating collaborations (Adams et al., 2005). Geographic proximity influences the frequency and persistence of collaborations, although technological advancements have alleviated some of the barriers associated with distance (Wu et al., 2024).

Research on scientific collaboration has identified a wide array of factors influencing its formation and persistence, including individual attributes, institutional and geographic considerations. These factors underscore the necessity for a unified modeling approach that can seamlessly integrate multi-source academic information. Such an approach aligns with the objective of this study, which leverages hypergraphs to maximize the utilization of all available information, thereby enhancing the effectiveness of collaborator recommendations.

Table 1
Node types in the scientific collaboration hypergraph.

Node type	Description	Example
Scholar (S)	Individual researchers	Alice, Bob
Gender (G)	Gender categories	Male, Female
Career stage (C)	Professional stages	Ph.D, Professor, Researcher
Expertise (E)	Research fields or topics	Machine learning, Bioinformatics
Institution (I)	Affiliated institutions	MIT, Peking University
Productivity (P)	Quantitative indicators	h-index, number of publications
Publication (A)	Academic publications	<i>Attention is All You Need</i>
Venue (V)	Journals or conferences	<i>Management Science</i>
Location (L)	Regions or countries	United States, China

2.3. Hypergraph representation learning

Hypergraph-based recommendation has gained increasing attention for its capacity to model complex many-to-many relations beyond the scope of traditional graphs. At the heart of these methods lies hypergraph representation learning, which embeds nodes and hyperedges into a latent space that preserves both structural and semantic dependencies (Antelmi et al., 2023). Early research in this area was predominantly rooted in spectral theory, where the hypergraph Laplacian is leveraged to extract global connectivity patterns via eigen-decomposition. Such spectral methods have demonstrated effectiveness in clustering and collaborative filtering tasks, where capturing the overall structure is crucial (Li et al., 2023; Zheng et al., 2018). Nonetheless, their reliance on costly matrix operations and limited adaptability to heterogeneous data significantly hampers scalability (Zhou et al., 2006).

To address these issues, researchers explored lighter proximity-based techniques that aim to preserve local topological structures, such as node similarity and hyperedge proximity (Jiang et al., 2020; Ma et al., 2018). These methods offer computational efficiency and simplicity, making them suitable for static and homogeneous recommendation tasks. However, their shallow architecture limits their capacity to capture non-linear semantics or cross-context dependencies, factors that are increasingly important in real-world recommendation scenarios.

More recently, deep learning-based hypergraph models have emerged as the dominant paradigm. Hypergraph convolutional network (HyperGCN), inspired by graph neural networks, enable message propagation across hyperedges, effectively integrating multi-hop and high-order relations (Yadati et al., 2019). Attention mechanisms further refine this learning process by dynamically adjusting the influence of neighboring nodes and hyperedges, enhancing predictive accuracy (Kim et al., 2020). These approaches have achieved impressive results in document, movie, and music recommendation (Gatta et al., 2023; Ji et al., 2020; Zhu et al., 2016).

Hypergraph learning has demonstrated substantial promise across multiple domains, yet its application to academic recommendation remains limited. Although a few studies have explored hypergraph-based modeling in academic contexts, such as using coauthorship hypergraphs for team formation analysis (Datta et al., 2014) or modeling author collaboration transitions with geometric hypergraphs (Xie et al., 2018), these works rely on handcrafted structures or heuristic assumptions. They do not incorporate heterogeneous attributes or end-to-end embedding mechanisms, which limits their scalability and effectiveness in large-scale scholarly environments. In summary, existing models tend to assume homogeneous relations and overlook the layered complexity of real-world collaboration, including differences in career stages, institutional affiliations, and research expertise. Recent advances in contrastive and generative learning for graph-based recommendation (Cai et al., 2025; Li et al., 2025; Xia et al., 2022; Zhang et al., 2025; Zhu et al., 2016) further underscore the need for structurally expressive and semantically robust modeling strategies in complex recommendation tasks.

3. Methodology

In this section, we define the hypergraph structure and the problem of scientific collaborator recommendation, design the proposed recommendation model and interpretability mechanism, and outline the experimental setups.

3.1. Problem formulation

3.1.1. Hypergraph construction

Modeling the multifaceted nature of academic collaboration requires a structure capable of representing complex interactions. We construct a heterogeneous hypergraph $\mathcal{H}$ in which nodes represent diverse entities and hyperedges capture high-order relationships among them. This structure enables the integration of both individual attributes and contextual factors into a unified representation.

As discussed in Section 2.2, we conducted a comprehensive literature review of influential studies on academic collaboration, identifying core factors. Table 1 summarizes the node types used in our model, including their semantic roles and examples. Table 2 outlines the hyperedge types, covering both explicit collaboration events and auxiliary academic relationships. Together, they form the structural backbone of the SCH, enabling expressive and interpretable representation learning over complex academic networks.

Formally, we define the scientific collaboration hypergraph as:

$$\mathcal{H} = (\mathcal{V}, \mathcal{E} = \{e_i : i \in I\}) \quad (1)$$

Table 2
Hyperedge types in the scientific collaboration hypergraph.

Hyperedge type	Connected node types	Description
Coauthorship	$S - A$	Links scholars and their joint publications
Citation link	$A - A$	Connects citing and cited publications
Gender association	$S - G$	Represents the gender of a scholar
Career progression	$S - C$	Indicates a scholar's professional stage
Research interest	$S - E$	Reflects research topics of scholars
Institutional affiliation	$S - I$	Links scholars to their institutions
Venue association	$A - V$	Connects publications to publishing venues
Geographic proximity	$S - L$	Connects scholars to their locations
Impact link	$S - P$	Connects scholars to productivity metrics



Fig. 2. A visual comparison of different coauthorship modeling strategies. Dashed boxes represent publication nodes used to induce projected coauthorship links.

where $\mathcal{V}$ is the node set, and $\mathcal{E}$ is the set of hyperedges indexed by a finite set I .

Specifically, the node set $\mathcal{V}$ is defined as:

$$\mathcal{V} = S \cup G \cup C \cup R \cup I \cup A \cup P \cup V \cup L \quad (2)$$

where each symbol corresponds to a node type as defined in Table 1.

Each hyperedge $e_i \in \mathcal{E}$ is represented as a tuple:

$$e_i = (e_i^+ = (\mathcal{V}_{e_i}^+, i); e_i^- = (i, \mathcal{V}_{e_i}^-)) \quad (3)$$

where e_i^+ and e_i^- denote the head and tail of the hyperedge. Here, $\mathcal{V}_{e_i}^+ \subseteq \mathcal{V}$ and $\mathcal{V}_{e_i}^- \subseteq \mathcal{V}$ represent the sets of nodes connected as inputs and outputs in the hyperedge, respectively. The complete set of participating nodes is $\mathcal{V}_{e_i} = \mathcal{V}_{e_i}^+ \cup \mathcal{V}_{e_i}^-$.

3.1.2. Collaborator recommendation task

Given the heterogeneous scientific collaboration hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ defined in Section 3.1.1, the collaborator recommendation task aims to predict potential hyperedges that are currently unobserved. Specifically, for a given scholar node $s \in S \subseteq \mathcal{V}$, the objective is to identify a ranked list of candidate scholars $\hat{S}_k \subseteq S$ who are most likely to form new coauthorship hyperedges with s .

In the SCH, coauthorship is modeled as a multi-way hyperedge connecting a set of scholar nodes with one or more publication nodes. While this structure captures the full semantics of collaboration events, our recommendation task ultimately requires predicting pairwise coauthorship likelihoods between scholars.

To bridge this representational gap, we project each coauthorship hyperedge that links a group of scholars $\{s_1, s_2, \dots, s_k\}$ with a set of publications A into a collection of scholar-scholar links $\{(s_i, s_j) \mid i \neq j\}$. This projection yields an induced coauthorship graph among scholars, which allows us to formulate the recommendation task as a standard link prediction problem over scholar pairs. Fig. 2 provides a visual comparison of different coauthorship modeling strategies, including our projection-based approach.

Formally, let $N(s)$ denote the set of scholars already connected to s via coauthorship. The recommendation task is to identify a subset of candidates $s' \in S \setminus N(s)$ who are most likely to collaborate with s in the future, based on a scoring function over hypergraph embeddings. The details of this scoring function, training objective, and inference mechanism are presented in Section 3.2.3.

3.2. Model design

3.2.1. Overview of the architecture

To address the collaborator recommendation task, we propose a hypergraph-based neural framework that operates directly on the Scientific Collaboration Hypergraph. As illustrated in Fig. 3, the overall architecture consists of three sequential stages: (1) **hypergraph representation learning**, (2) **coauthorship likelihood modeling**, and (3) **interpretability mechanism**.

Each stage transforms the outputs of the previous stage: the first stage maps SCH into scholar embeddings; the second stage translates these embeddings into recommendation scores and top- k recommendations; and the third stage maps these predictions into human-readable explanations. This modular architecture enables both effective performance and transparent interpretability.

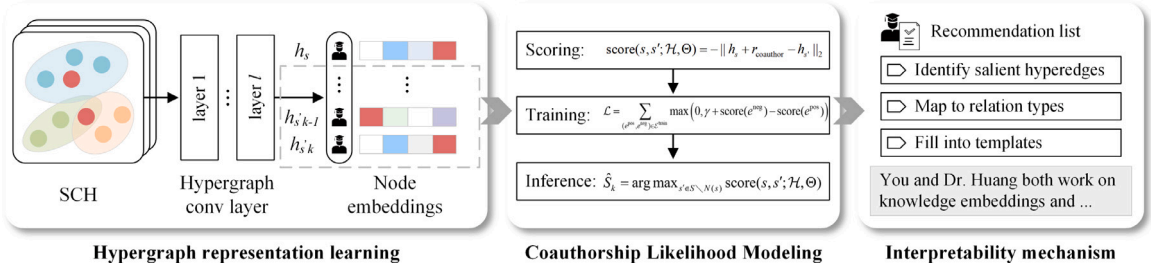


Fig. 3. The overall architecture of our proposed framework. The gray dashed box in the first stage indicates the *candidate collaborator pool* whose embeddings will be evaluated.

3.2.2. Hypergraph representation learning

The first stage of our framework aims to learn node embeddings from the heterogeneous SCH. Modeling such multi-entity collaborations requires a method capable of capturing global structural patterns. To this end, we adopt a spectral hypergraph neural network, which propagates signals based on the eigenstructure of the hypergraph Laplacian.

Each node is initially assigned a one-hot identity vector $\mathbf{x}_v \in \mathbb{R}^{|\mathcal{V}|}$ to ensure uniqueness. These vectors are subsequently projected into a latent space of dimension d' via the first hypergraph convolutional layer, yielding structure-aware embeddings $\mathbf{h}_v \in \mathbb{R}^{d'}$.

To propagate information across the hypergraph, we employ a sparsity-preserving message passing scheme inspired by HyperGCN (Yadati et al., 2019). For each hyperedge e_i , instead of constructing a full clique among all nodes in e_i , we approximate it by connecting each node to its most structurally representative neighbor within the same hyperedge. This reduces the number of pairwise connections from $O(|e_i|^2)$ to $O(|e_i|)$, enabling scalable learning while maintaining core structural signals.

We mitigate over-smoothing by restricting the network depth to two convolutional layers and incorporate non-linear activation (ReLU) after each layer to maintain representational diversity. Furthermore, we apply attention-based normalization to dynamically weigh the influence of neighboring nodes and hyperedges, allowing the model to emphasize informative collaboration contexts while suppressing noise, which is an effective strategy under sparse data conditions.

The embedding of node v at layer $l + 1$ is computed as:

$$\mathbf{h}_v^{(l+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} \bar{A}_{vu}^{(l)} \mathbf{h}_u^{(l)} \mathbf{W}^{(l)} \right) \quad (4)$$

where $\mathcal{N}(v)$ denotes the neighborhood of node v induced by the approximated edge set, $\bar{A}^{(l)}$ is the normalized adjacency matrix, $\mathbf{W}^{(l)}$ is a learnable weight matrix, and $\sigma(\cdot)$ is a non-linear activation function.

After L layers, we obtain the final node embeddings $\mathbf{H} = \{\mathbf{h}_v^{(L)} \mid v \in \mathcal{V}\}$, which serve as input to the scoring module in the next stage.

While this strategy may omit certain fine-grained interactions, we mitigate potential limitations in two ways. First, attention-based normalization in the adjacency matrix $\bar{A}^{(l)}$ adaptively weighs neighbor contributions, alleviating over-smoothing and enabling the model to prioritize informative signals. Second, the use of typed hyperedges (e.g., coaffiliation, shared expertise) preserves semantic heterogeneity and mitigates sparsity by providing multiple interaction contexts.

3.2.3. Coauthorship likelihood modeling

While HyperGCN provides an effective means of learning hypergraph-based node embeddings, it is primarily designed for semi-supervised node classification tasks, where the objective is to predict a label for each individual node. However, in our setting, the goal is to predict potential coauthorships between pairs of scholar nodes, which requires a fundamentally different downstream formulation. Therefore, we design a coauthorship likelihood modeling module that operates on the embeddings produced by the encoder and estimates the probability of collaboration between a query scholar and each candidate.

Scoring function. Given the learned embeddings $\mathbf{h}_s$ and $\mathbf{h}_{s'}$ for the target scholar s and a candidate collaborator s' , we define a translational scoring function inspired by knowledge graph embedding methods (Bordes et al., 2013) to measure their coauthorship compatibility:

$$\text{score}(s, s'; \mathbf{H}, \Theta) = -\|\mathbf{h}_s + \mathbf{r}_{\text{coauthor}} - \mathbf{h}_{s'}\|_2 \quad (5)$$

where $\mathbf{r}_{\text{coauthor}} \in \mathbb{R}^{d'}$ is a trainable relation vector representing the semantic translation of a coauthorship relation in the embedding space.

Training objective. Model parameters are optimized using a pairwise ranking loss, which favors higher scores for true collaborators over randomly sampled negative candidates. Given a set of positive and negative scholar pairs (s, s^+) and (s, s^-) , the loss is defined as:

$$\mathcal{L} = \sum_{(s, s^+, s^-)} \max(0, \gamma + \text{score}(s, s^-) - \text{score}(s, s^+)) \quad (6)$$

where $\gamma > 0$ is a margin hyperparameter controlling the separation between positive and negative samples.

Table 3
Node and hyperedge statistics of the scientific collaboration hypergraph.

Node	# entries	Hyperedge	# entries
Scholar (S)	6,119	Coauthorship ($S-A$)	18,092
Gender (G)	2	Citation link ($A-A$)	46,382
Career stage (C)	11	Gender association ($S-G$)	6,119
Expertise (E)	312	Career progression ($S-C$)	6,119
Institution (I)	961	Research interest ($S-E$)	39,841
Productivity (P)	5	Institutional affiliation ($S-I$)	6,119
Publication (A)	18,092	Venue association ($A-V$)	18,092
Venue (V)	538	Geographic proximity ($S-L$)	6,119
Location (L)	112	Impact link ($S-P$)	6,119

Note: Productivity levels (P) are quantized from h-index values; raw h-index values are preserved as node features.

Inference strategy. At inference time, we generate recommendations by ranking all candidate scholars based on their collaboration scores with the query scholar. Formally, the top- k recommended collaborators are obtained as:

$$\hat{S}_k = \arg \max_{s' \in S \setminus N(s)} \text{score}(s, s'; H, \Theta) \quad (7)$$

where $N(s)$ denotes the set of scholars who have already coauthored with s . The score reflects the estimated likelihood that s and s' will form a new coauthorship in the future.

3.2.4. Interpretability mechanism

A lightweight interpretability module is incorporated to improve the transparency of the recommendation system. Leveraging the multi-relational schema of the heterogeneous SCH, we derive a rich set of interpretable paths. Instead of treating the embedding space as a black box, our approach uses hyperedge types as an explicit *rule base* to uncover the structural and semantic factors underlying each recommendation.

For each recommended coauthor s' , we identify the most influential hyperedge types connecting s and s' by aggregating attention weights learned from the hypergraph convolution layers. These weights reflect the relative contribution of each hyperedge type to the node representation. Specifically, we compute the total attention weight for each hyperedge type and use these scores to rank their importance. These are then mapped to explanation templates:

- Coaffiliation: “Both scholars are affiliated with {institution}”.
- ResearchExpertise: “Both scholars work on {expertise}”.
- PublicationVenue: “Both scholars have published in {venue}”.

Given a top- k recommendation $\hat{S}_k$ for scholar s , we generate an explanation sentence for each $s' \in \hat{S}_k$ by retrieving the most salient hyperedge types connecting s and s' in the SCH, and instantiating the corresponding template.

4. Experimental setup

This section describes the dataset used to construct the SCH, the evaluation metrics used to measure recommendation performance, the baseline methods for comparison, and the implementation details of our method.

4.1. Dataset

We conduct our experiments on the AMiner academic dataset (Tang et al., 2008), a large-scale and heterogeneous repository encompassing publication metadata, authorship records, institutional affiliations, venues, and research topics. These sources instantiate our hypergraph: coauthorship is encoded as event-level hyperedges, and the other relations as their respective hyperedges.

To construct a reliable Scientific Collaboration Hypergraph, we aggregate data from multiple AMiner subsets. The Academic Social Network provides large-scale coauthorship and citation relations, while additional resources such as Web User Profiling (Gu et al., 2016) and Scholar Profiling (Zhang et al., 2024) supply essential scholar attributes including gender and career stage.

All scholar entities are drawn from the disambiguated author ID set provided by AMiner. To ensure structural consistency and semantic richness, we retain only those scholars with complete information across all required fields. This multi-source integration approach ensures data quality while preserving the high-order collaboration patterns necessary for effective hypergraph modeling. The resulting heterogeneous relations drive our spectral hypergraph and attention modules, supporting cold-start and interdisciplinary cases. Table 3 summarizes the number of nodes and hyperedges of each type in the SCH.

4.2. Evaluation metrics

We adopt two classes of evaluation metrics to comprehensively evaluate the proposed collaborator recommendation framework: *effectiveness* metrics to assess the quality of recommendations, and *efficiency* metrics to measure computational cost.

We evaluate recommendation *effectiveness* using two widely adopted ranking metrics: Recall@k and Normalized Discounted Cumulative Gain (NDCG@k), with $k \in \{10, 20, 50\}$. For a given scholar s , let I_s denote the set of held-out ground-truth collaborators and $\omega(k')$ denote the collaborator ranked at position k' by the model. Recall@k is defined as:

$$\text{Recall}@k(s, \omega) = \frac{1}{\min(k, |I_s|)} \sum_{k'=1}^k \mathbb{I}[\omega(k') \in I_s] \quad (8)$$

In addition, we compute the Discounted Cumulative Gain (DCG@k), which accounts for the rank positions of relevant collaborators, as follows:

$$\text{DCG}@k(s, \omega) = \sum_{k'=1}^k \frac{2^{\mathbb{I}[\omega(k') \in I_s]} - 1}{\log_2(k' + 1)} \quad (9)$$

NDCG@k is obtained by dividing DCG@k by the ideal DCG, which assumes all relevant collaborators are ranked at the top, thereby normalizing the score to the range [0, 1].

In terms of *efficiency*, we report two metrics: training time, the total runtime required to train the model; and inference time, the average time required to generate a top-k recommendation list for a single scholar.

4.3. Baseline comparisons

To benchmark the performance of our proposed collaborator recommendation framework, we compare it against several strong baselines drawn from recent advances in academic recommendations. These include: Mult-VAE, NGCF, LightGCN, GF-CF, and OAG-BERT.

While recent large language models (LLMs), such as GPT-based architectures, have shown strong performance in general-purpose NLP tasks, they are not directly included as baselines in this study. Our task models explicit multi-entity collaboration events with heterogeneous hyperedges, and serializing these structures into linear prompts risks losing event semantics, which weakens controllability and comparability. Using LLMs typically requires substantial prompt design or fine tuning with task specific data, introducing reproducibility concerns and evaluation inconsistency (Bellogin & Said, 2021; Errica et al., 2025). Large scale inference also adds significant computational and data governance burdens, complicating standardized offline evaluation. We therefore adopt OAG-BERT as a strong, task aligned pretrained transformer for scholarly recommendation (Liu et al., 2022). It runs locally and yields deterministic outputs under fixed configurations, supporting consistent and reproducible experiments.

4.4. Implementation details

We provide a detailed account of the experimental configuration, covering the training environment, model hyperparameters, and implementation strategies for the baseline methods. All settings are aligned to ensure the comparability and reproducibility of results.

Training environment. All experiments are conducted on a personal workstation equipped with an Intel Core i7-13700KF CPU, 32 GB RAM, and an NVIDIA GeForce RTX 3080 GPU with 10 GB VRAM. The implementation is based on PyTorch, with hypergraph operations implemented using custom modules.

Hyperparameters. We set the embedding dimension $d' = 128$ and use a two-layer hypergraph convolutional encoder. The model is trained using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 512. Early stopping is applied based on validation Recall, with a patience of 10 epochs. The margin hyperparameter γ for the pairwise ranking loss is set to 1.0 (see Appendix A for hyperparameter sensitivity analysis).

Implementation. For reproducibility, we adopt publicly available and well-maintained implementations for all baseline models. When multiple variants exist, we select the most widely used and best-performing version to ensure fair and representative comparisons. All baselines are trained using their original recommended hyperparameters unless otherwise specified.

- **Mult-VAE:** Implemented in TensorFlow based on the official codebase by Liang et al. (2018), with adaptations for coauthorship-style implicit feedback.
- **NGCF and LightGCN:** Both implemented in TensorFlow using the official repositories by He et al. (2020) and Wang et al. (2019). Coauthorship graphs are constructed as undirected user-item interactions.
- **GF-CF:** Implemented in PyTorch following the official repository of Shen et al. (2021), with spectral graph filtering modules adapted for coauthorship links.
- **OAG-BERT:** Following the released PyTorch implementation from Liu et al. (2022), we fine-tune the pretrained model on our SCH dataset using authors, publication titles, and abstracts as input. Final scholar embeddings are obtained from the [CLS] token representations.

All baselines share the same data splits and supervision signals as our model. Coauthorship edges are used to construct graphs for all graph-based baselines, and identical metadata is provided to all semantic models for fairness.

Table 4

Recommendation *effectiveness* comparison. Bold indicates the best method, *italic* indicates the second-best method.

Method	Recall@k			NDCG@k		
	@10	@20	@50	@10	@20	@50
Mult-VAE	0.0104	0.0171	0.0258	0.0073	0.0120	0.0185
NGCF	0.0159	0.0246	0.0337	0.0111	0.0175	0.0249
LightGCN	0.0186	0.0298	0.0415	0.0137	0.0209	0.0292
GF-CF	0.0214	0.0335	0.0462	0.0154	0.0238	0.0325
OAG-BERT	<i>0.1013</i>	<i>0.1305</i>	<i>0.1772</i>	<i>0.0540</i>	<i>0.0841</i>	<i>0.1220</i>
Ours	0.1802*	0.2236*	0.2693*	0.1315*	0.1627*	0.2014*

Note:

* Denotes statistically significant improvement over the second-best method at $p < 0.001$.

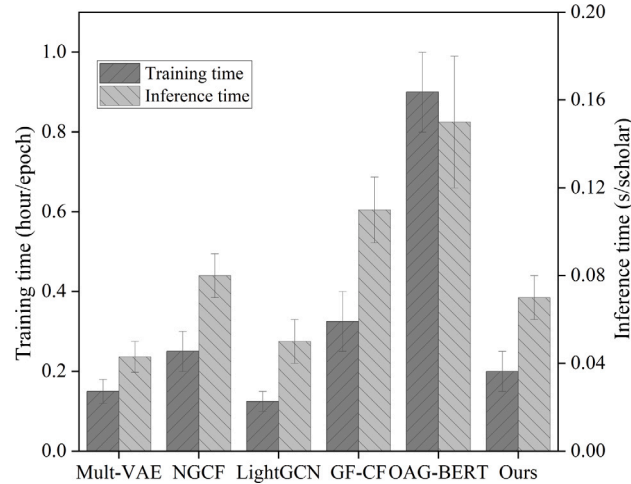


Fig. 4. Recommendation efficiency comparison. *mean* $\pm$ 95% confidence interval.

5. Experimental results

5.1. Overall performance comparison

5.1.1. Recommendation effectiveness

Table 4 reports the overall recommendation effectiveness of all baselines. Our proposed method achieves the best performance across all settings, consistently outperforming both traditional collaborative filtering models and recent neural and pretrained baselines.

Notably, our approach achieves a Recall@10 of 0.1802, representing a relative improvement of over 78% compared to the second-best model OAG-BERT (0.1013). A similar trend holds for NDCG@10, where our method achieves 0.1315, substantially surpassing OAG-BERT's 0.0540. The performance advantage remains consistent as k increases, suggesting that our model effectively captures both top-ranked and broad collaborative signals.

5.1.2. Recommendation efficiency

We begin our evaluation on recommendation *efficiency* by comparing the computational efficiency of all methods. As shown in Fig. 4, we report both the per-epoch training time (in hours) and the average inference time per scholar (in seconds), measured under a consistent hardware setup. Error bars indicate the 95% confidence intervals.

As shown in the figure, our method maintains competitive training efficiency, requiring only 0.2 h/epoch, which is notably faster than GF-CF (0.325) and OAG-BERT (0.9), while slightly exceeding the lightweight Mult-VAE (0.15). In terms of inference, our method completes top- k recommendation generation in approximately 0.07 s, significantly outperforming OAG-BERT (0.15) and GF-CF (0.11), and aligning closely with LightGCN (0.05). Overall, the results demonstrate that our model achieves a favorable balance between computational efficiency and recommendation accuracy, making it well-suited for large-scale academic applications.

5.1.3. Cold-start analysis

Robustness under limited supervision is crucial for practical recommendation systems. To examine this, we evaluate model *effectiveness* in cold-start scenarios, where scholars in the training set have fewer than five historical collaborations.

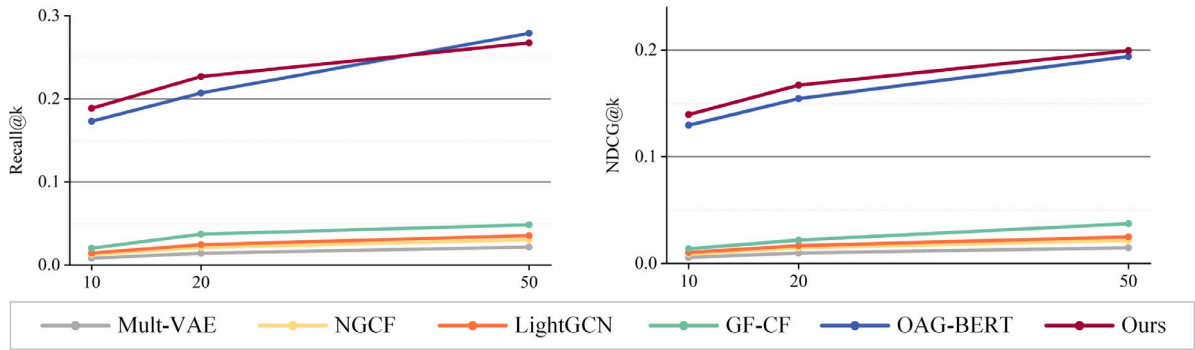


Fig. 5. Cold-start experiment results. Recall@k and NDCG@k scores on test set scholars with five or fewer collaborators in the corresponding training set.

As illustrated in Fig. 5, our method consistently outperforms all baselines across different cutoff values. Notably, at $k = 10$, the model achieves even higher recall (0.1886) and NDCG (0.1395) compared to the full test set (0.1802 and 0.1315, respectively). This performance gain can be attributed to three factors. First, our model leverages rich high-order structures in SCH that can effectively compensate for the lack of collaboration history. Second, cold-start scholars often have more specific and less noisy semantic profiles, allowing the model to generate sharper recommendations. Third, since cold-start scholars typically have fewer ground-truth collaborators, successfully recovering even one or two leads to proportionally higher recall and NDCG scores.

These results indicate that our hypergraph-based approach can effectively generalize beyond existing collaboration history by leveraging rich information in SCH. More importantly, it demonstrates a strong capability to recommend collaborators for early-career or sparsely connected scholars, helping them break out of narrow academic circles and discover potential collaborators.

5.2. Embedding representation

To provide a more intuitive understanding of the learned embedding space, we project the final node embeddings of scholars into two dimensions using t-SNE (van der Maaten & Hinton, 2008), as shown in Fig. 6. Since the dataset spans a wide range of research expertise categories, displaying all of them would lead to a cluttered visualization. To address this, we select the five most common expertise types in the dataset: *case study*, *neural network*, *genetic algorithm*, *sensor network*, and *data mining*. These categories are marked with distinct colors, while all remaining expertise types are grouped together and displayed in a default color.

As illustrated in the figure, the learned embedding space does not form isolated clusters strictly along research domains. Instead, scholars from different fields are distributed across shared regions, suggesting that the model captures latent collaboration structures that transcend disciplinary boundaries. For example, the area around Michael E. Houle and Kaiming He shows noticeable overlap between scholars working on *neural networks*, *genetic algorithms*, and *data mining*, indicating a shared methodological foundation. Similarly, the neighborhood surrounding Yang Richard Yang exhibits a mix of *sensor network* and *neural network* scholars, reflecting the interdisciplinary nature of applied research in networking and systems. Rather than segregating scholars by field, the embedding reveals areas of potential intellectual convergence, which is particularly valuable for identifying novel and interdisciplinary collaboration opportunities.

Further, we complement the qualitative embedding visualization with a quantitative assessment of interdisciplinarity. To this end, we adopt a widely used metric in scientometric studies known as disciplinary entropy, which measures the diversity of each scholar's collaboration profile (Porter et al., 2007). This metric captures the extent to which a scholar's coauthors are distributed across different research domains. A higher entropy score indicates a greater degree of interdisciplinary collaboration.

Specifically, we evaluate whether the learned embeddings reflect real-world interdisciplinary collaboration patterns by comparing them against these entropy scores. For each scholar, we compute an embedding-based diversity score by identifying their k nearest neighbors in the embedding space ($k = 10$) and calculating the proportion of neighbors belonging to different research domains. This local diversity score quantifies how heterogeneous a scholar's semantic neighborhood is with respect to disciplinary affiliation. We then compute the Pearson correlation between each scholar's embedding-based diversity score and their coauthor entropy score. The result is $r = 0.87$ ($p < 0.001$). The strong correlation between embedding-based diversity and actual collaboration entropy suggests that the learned embeddings reflect real-world interdisciplinary structures.

5.3. Interpretability evaluation

Evaluating the interpretability of recommendation systems from a human-centered perspective is of critical importance (Gao et al., 2023). We conduct a user study involving 150 participants from diverse academic backgrounds and career stages, recruited from Peking University, Beijing Jiaotong University, Beijing University of Chinese Medicine, and other institutions. Each participant interacts with our collaborator recommendation system and receives personalized recommendations with corresponding

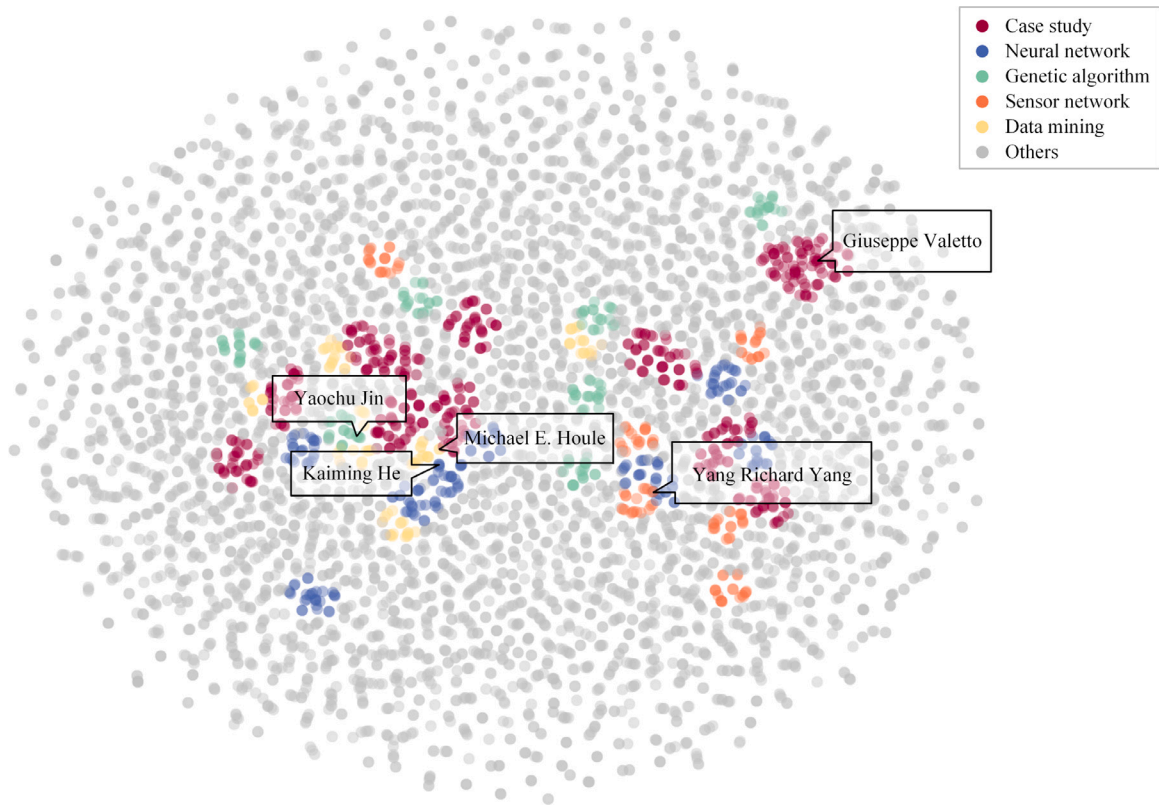


Fig. 6. Visualization of learned scholar embeddings using t-SNE. Color indicates distinct research domains. Several scholars from five domains are annotated to facilitate intuitive understanding.

Table 5

Descriptive statistics for explanation quality dimensions. All ratings are based on 5-point Likert scales.

Dimension	Mean	Std. Dev.
Comprehensibility	3.8133	0.9159
Trust	3.9600	0.8574
Usefulness	4.0400	0.9506
Transparency	3.7111	0.8996
Satisfaction	3.6200	1.0555

natural language explanations derived from hyperedge semantics. After the interaction session, participants complete a structured questionnaire.

Following the guidelines proposed by [Zhang and Chen \(2020\)](#), we assess user perceptions of the explanation module across five key dimensions: *Comprehensibility*, *Trust*, *Usefulness*, *Transparency*, and *Overall Satisfaction*. These dimensions are widely recognized in the literature as the core user-centered goals of explainable recommendation systems. Each dimension is measured using a 3-item, 5-point Likert subscale (see [Appendix B](#) for detailed questionnaire design, along with reliability and validity analysis).

To quantitatively assess the interpretability quality of our system, we computed descriptive statistics for the five explanation quality dimensions. [Table 5](#) summarizes the mean scores and standard deviations across all participants. Overall, the scores indicate positive user perceptions ([Kouki et al., 2019](#)), with all dimensions averaging above 3.6 out of 5. Notably, the *Usefulness* and *Trust* dimensions received the highest average ratings ($M = 4.04$ and 3.96 , respectively), suggesting that users found the explanations both helpful and trustworthy.

5.4. Case study

To further demonstrate the practical value and interpretability of our system, we conduct a case study based on a real-world early-career scholar who is currently pursuing a doctoral degree. The system takes the user's academic profile as input and generates

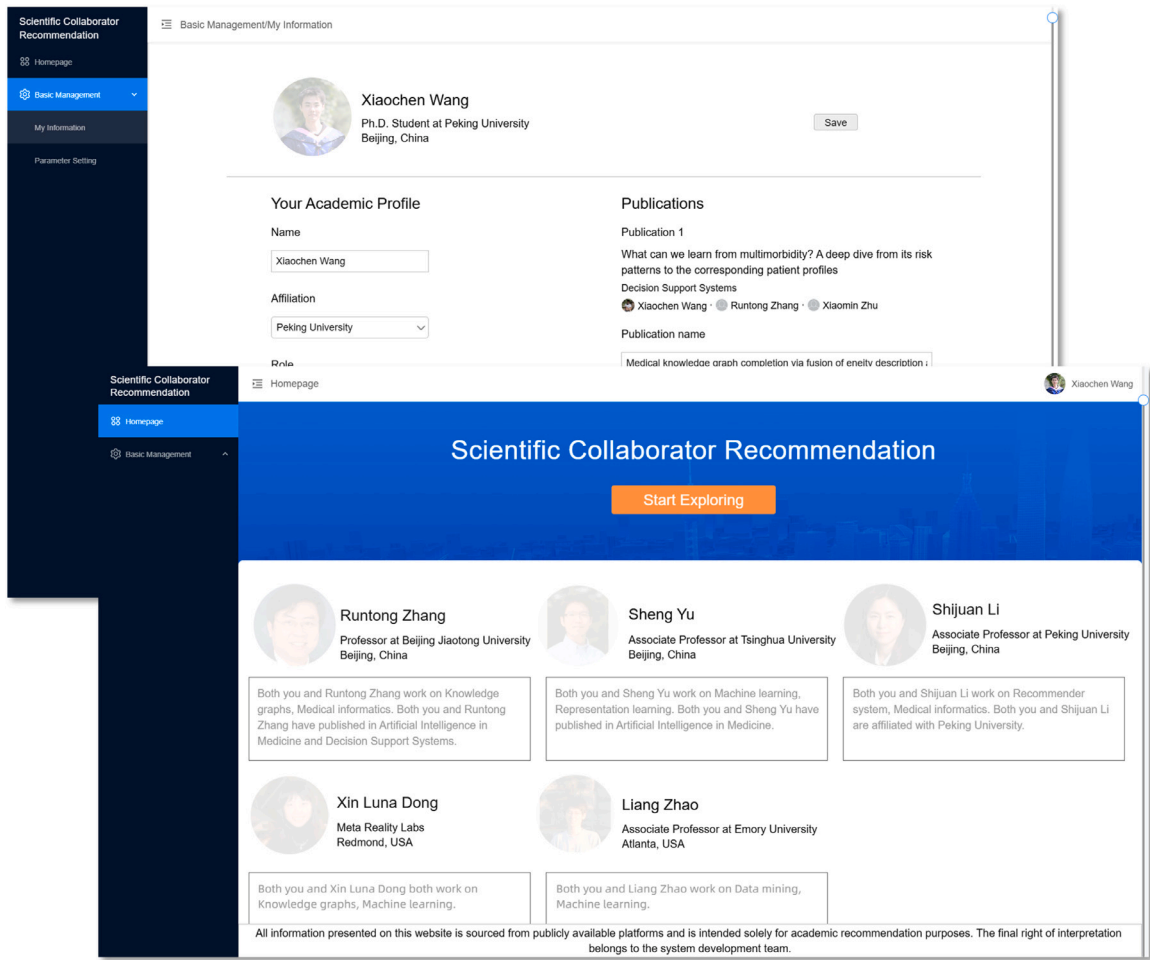


Fig. 7. An illustration of case study. Collaborator recommendation results for Xiaochen Wang, including academic profile input (top) and system-generated top-5 collaborator recommendations with hyperedge-based natural language explanations (bottom).

a ranked list of recommended collaborators. For each recommendation, the system provides a natural language explanation, constructed from the semantic relations embedded in hyperedges.

The recommendation process begins with the user completing their academic profile through the “My Information” module. Upon saving, the backend automatically updates the SCH to reflect the newly entered information. In the “Parameter Setting” module, the user then specifies the number of desired recommendations. The system subsequently performs hypergraph embedding and generates a ranked list of suitable collaborators.

As illustrated in Fig. 7, the top-5 collaborators recommended for Xiaochen Wang include both domestic and international scholars such as Sheng Yu (Tsinghua University) and Xin Luna Dong (Meta Reality Labs). The system-generated explanations highlight shared research domains (e.g., knowledge graphs) and co-publication venues (e.g., *Decision Support Systems*). These examples demonstrate how semantic hyperedges effectively support interpretable and personalized academic recommendations.

6. Conclusions and future research

General discussion. This study presents a novel hypergraph-based framework for scientific collaborator recommendation. Unlike prior approaches that rely on pairwise coauthorship networks or textual similarity, our model introduces a heterogeneous Scientific Collaboration Hypergraph to explicitly encode high-order, multi-relational academic structures. This structural design allows the model to uncover complex collaborative patterns that are often lost in traditional graph or embedding-based methods. In addition, our interpretable recommendation module bridges the gap between accuracy and transparency, distinguishing this study from prior work that treats recommendation as a black-box process.

Main findings. Extensive experiments demonstrate that the proposed model outperforms state-of-the-art baselines across multiple evaluation metrics, including Recall and NDCG. Performance gains are particularly notable in cold-start scenarios and interdisciplinary recommendations. Embedding visualization confirms the semantic alignment of learned representations with real-world collaboration profiles, while a user study validates the interpretability and perceived utility of the recommendation outputs.

Theoretical implications. Encoding collaboration as event-level hyperedges preserves group semantics and avoids the information loss of dyadic decomposition. A type-aware spectral hypergraph convolution with a representative-neighbor approximation reduces induced edges while maintaining the principal structural signal. Adapting translational distance scoring to the hypergraph embedding space yields a clear margin between observed and corrupted collaboration configurations. Attention over relation types provides intrinsic contribution scores that can be surfaced as natural-language rationales, creating a direct link between representation learning and interpretability.

Practical implications. The proposed system is well suited to settings with sparse histories or exploratory collaboration needs, including support for early-career researchers and the assembly of interdisciplinary teams. It integrates with academic platforms in batch ranking or interactive assistant modes, where relation-level rationales improve transparency, auditability, and user trust. The ability to run with fixed configurations and deterministic outputs also facilitates governance and reproducible evaluation in production environments.

Limitations and future work. Despite these contributions, several limitations remain. First, our experiments are conducted on a filtered subset of AMiner, which, while representative, may not capture the full diversity of global academic networks. Second, the explanation module is rule-based and may lack the flexibility of generative models. Future work could explore retrieval-augmented generation or hypergraph-informed LLMs to improve personalization. Third, we did not include a dedicated comparison with knowledge graph recommenders; we plan a systematic KG baseline study and will explore hybrid designs that combine event level hyperedges with triple level reasoning. Finally, we recognize the computational overhead associated with hypergraph embedding and plan to investigate scalable variants or distillation-based optimizations for real-world deployment.

CRediT authorship contribution statement

Xiaochen Wang: Writing – original draft, Writing – review & editing, Visualization, Software, Methodology, Formal analysis. **Wensheng Huang:** Data curation. **Butian Zhao:** Validation. **Shijuan Li:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Social Science Foundation of China (23BTQ018), the National Natural Science Foundation of China (72474009), and the Fundamental Research Funds for the Central Universities, China (2025-JYB-XJSJJ-022). The authors thank Professor Yi Bu for his valuable suggestions.

Appendix A. Hyperparameter sensitivity analysis

A.1. Embedding dimension

We evaluated the impact of the embedding dimension d' on the performance and efficiency of our collaborator recommendation model. A higher d' allows the model to encode richer semantic and structural features, potentially improving recommendation quality. However, it also increases computational cost.

As shown in Fig. A.1, both Recall@ k and NDCG@ k improved substantially when increasing d' from 32 to 128, indicating that lower-dimensional embeddings may underfit the high-order relations encoded in the hypergraph. However, the performance gain began to saturate as d' increased from 128 to 256, with only marginal improvements, suggesting diminishing returns. Meanwhile, execution time grew considerably with increasing d' , from 1 h and 5 min at $d' = 32$ to 2 h and 52 mins at $d' = 256$. Notably, the jump from $d' = 128$ to $d' = 256$ added over one hour of execution time while offering minimal performance benefit. Taken together, these results indicate that an embedding dimension of 128 provides a desirable balance between effectiveness and efficiency, and is thus adopted as the default setting in our model.

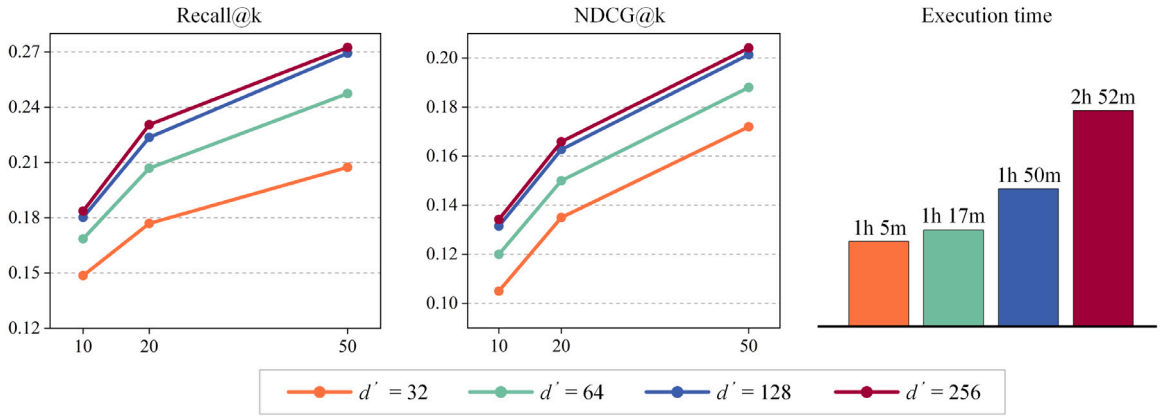


Fig. A.1. Impact of embedding dimension d' on Recall@ k , NDCG@ k , and execution time required to generate embeddings. $d' \in \{32, 64, 128, 256\}$.

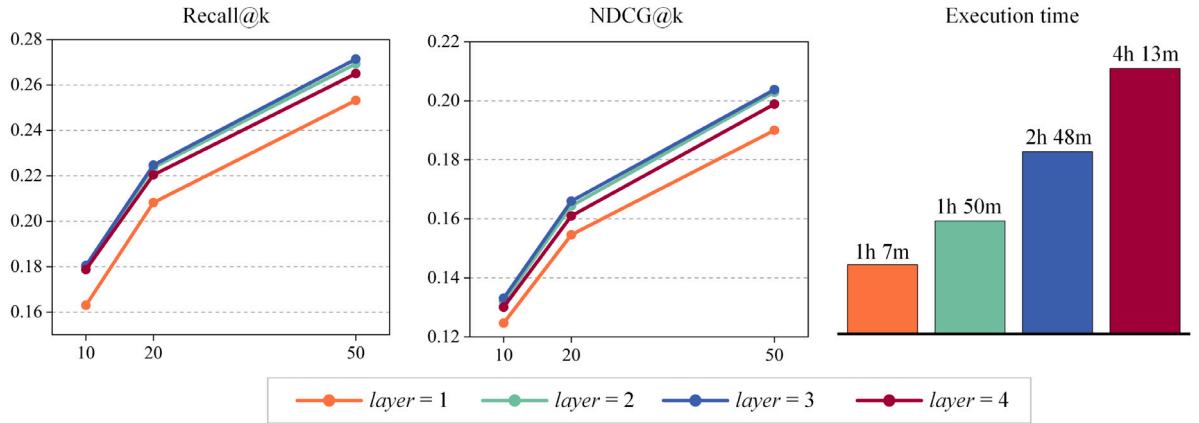


Fig. A.2. Impact of Hypergraph convolution layers on Recall@ k , NDCG@ k , and execution time required to generate embeddings. $layer \in \{1, 2, 3, 4\}$.

A.2. Hypergraph convolution layers

We also analyze the effect of the number of hypergraph convolution layers in the encoder. A deeper encoder enables the model to capture more complex high-order dependencies among nodes. We vary the number of layers from 1 to 4. As shown in Fig. A.2, using two layers consistently yields the best performance, outperforming both shallower (1-layer) and deeper (3- or 4-layer) configurations. Increasing the depth beyond two layers leads to marginal or even negative gains, potentially due to oversmoothing or parameter redundancy. This observation aligns with prior findings in hypergraph representation learning, where moderately deep architectures are typically sufficient to capture higher-order collaborative structures without incurring excessive computational overhead or performance degradation (Antelmi et al., 2023).

Appendix B. User study

B.1. User study questionnaire

This section provides the detailed design of the user study conducted to evaluate the quality of interpretability module in our collaborator recommendation system. The actual questionnaire version can be accessed at [Wenjuanxing\(inChinese\)](#). Table B.1 provides a comprehensive overview of the questionnaire dimensions and their corresponding questions.

B.2. Reliability and validity analysis

To evaluate the reliability of the questionnaire, we calculated Cronbach's alpha for each dimension. As shown in Table B.2, all dimensions exhibit acceptable to excellent reliability (all values > 0.7), indicating high internal consistency.

Table B.1

User study questionnaire structure.

User background	Gender (male/female)
	Career stage: undergraduate, postgraduate, etc (multiple choice).
	Research field (free-text)
	Have you previously used or sought collaborator recommendations? (yes/no)
Comprehensibility	I can clearly understand why the system recommended this collaborator.
	The content in the explanation made the rationale clear to me.
	The explanation used familiar terms or concepts that were easy to follow.
Trust	The explanation increased my trust in the system's recommendation.
	Without explanation, I would trust the system less.
	I believe the recommendations are more reliable when explanations are provided.
Usefulness	The explanation was helpful in judging whether the collaborator was suitable.
	The explanation revealed relevant information I might have overlooked.
	The explanation guided my decision to consider this collaborator.
Transparency	I feel that the recommendation logic is transparent rather than a black-box.
	The explanation helped me understand how the system works.
	The system's behavior feels more predictable when explanations are shown.
Satisfaction	Overall, I am satisfied with the explanation module.
	I found the explanation module to be a valuable part of the system as a whole.
	I would like to use similar explainable recommendation systems in the future.

Table B.2

Reliability test results for each dimension.

Dimension	Cronbach's alpha	Std. alpha	Items
Comprehensibility	0.806	0.808	3
Trust	0.772	0.772	3
Usefulness	0.846	0.847	3
Transparency	0.810	0.811	3
Satisfaction	0.846	0.848	3
Overall	0.875	0.875	15

Construct validity was examined using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's test of sphericity. The KMO value was 0.834, indicating meritorious sampling adequacy. Bartlett's test yielded a chi-square value of 1033.569 (df = 105, $p < .001$), suggesting that the variables are sufficiently correlated to conduct factor analysis.

Data availability

The data that support the findings of this study are openly available in AMiner at <http://doi.org/10.1145/1401890.1402008>.

References

- Adams, J. D., Black, G. C., Clemmons, J. R., & Stephan, P. E. (2005). Scientific teams and institutional collaborations: evidence from u.s. Universities, 1981–1999. *Research Policy*, 34(3), 259–285. <http://dx.doi.org/10.1016/j.respol.2005.01.014>.
- Antelmi, A., Cordasco, G., Polato, M., Scarano, V., Spagnuolo, C., & Yang, D. (2023). A survey on hypergraph representation learning. *ACM Computing Surveys*, 56(1), 24:1–24:38. <http://dx.doi.org/10.1145/3605776>.
- Bellogín, A., & Said, A. (2021). Improving accountability in recommender systems research through reproducibility. *User Modeling and User-Adapted Interaction*, 31(5), 941–977. <http://dx.doi.org/10.1007/s11257-021-09302-x>.
- Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., & Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. vol. 26, In *Advances in neural information processing systems*. Curran Associates, Inc.
- Cai, T., Zhang, Y., Li, K., Li, X., & Wang, X. (2025). Feature-decorrelation adaptive contrastive learning for knowledge-aware recommendation. *Neural Networks*, 190, Article 107646. <http://dx.doi.org/10.1016/j.neunet.2025.107646>.
- Datta, A., Yong, J. T. T., & Braghin, S. (2014). The zen of multidisciplinary team recommendation. *Journal of the Association for Information Science and Technology*, 65(12), 2518–2533. <http://dx.doi.org/10.1002/asi.23139>.
- Ebadi, A., & Schiffauerova, A. (2015). How to become an important player in scientific collaboration networks? *Journal of Informetrics*, 9(4), 809–825. <http://dx.doi.org/10.1016/j.joi.2015.08.002>.
- Errica, F., Siracusano, G., Sanvito, D., & Bifulco, R. (2025). What did i do wrong? Quantifying llms' sensitivity and consistency to prompt engineering. In *Proceedings of the 2025 annual conference of the North American chapter of the association for computational linguistics: human language technologies*.
- Gao, J., Peng, P., Lu, F., Claramunt, C., & Xu, Y. (2023). Towards travel recommendation interpretability: Disentangling tourist decision-making process via knowledge graph. *Information Processing & Management*, 60(4), Article 103369. <http://dx.doi.org/10.1016/j.ipm.2023.103369>.
- Gatta, V. L., Moscato, V., Pennone, M., Postiglione, M., & Sperli, G. (2023). Music recommendation via hypergraph embedding. *IEEE Transactions on Neural Networks and Learning Systems*, 34(10), 7887–7899. <http://dx.doi.org/10.1109/TNNLS.2022.3146968>.
- Gu, X., Yang, H., Tang, J., & Zhang, J. (2016). Web user profiling using data redundancy. In *2016 IEEE/ACM international conference on advances in social networks analysis and mining (ASONAM)* (pp. 358–365). <http://dx.doi.org/10.1109/ASONAM.2016.7752259>.

- Hara, N., Solomon, P., Kim, S.-L., & Sonnenwald, D. H. (2003). An emerging view of scientific collaboration: Scientists' perspectives on collaboration and factors that impact collaboration. *Journal of the Association for Information Science and Technology*, 54(10), 952–965. <http://dx.doi.org/10.1002/asi.10291>.
- He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., & Wang, M. (2020). LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* (pp. 639–648). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3397271.3401063>.
- He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T.-S. (2017). Neural collaborative filtering. In *WWW '17, Proceedings of the 26th international conference on world wide web* (pp. 173–182). Republic and Canton of Geneva, CHE: International World Wide Web Conferences Steering Committee, <http://dx.doi.org/10.1145/3038912.3052569>.
- He, C., Wu, J., & Zhang, Q. (2022). Proximity-aware research leadership recommendation in research collaboration via deep neural networks. *Journal of the Association for Information Science and Technology*, 73(1), 70–89. <http://dx.doi.org/10.1002/asi.24546>.
- Hill, N. S., & Hincapie, M. X. (2025). Gender differences in virtual collaboration effectiveness in hybrid teams. *Information & Management*, 62(3), Article 104107. <http://dx.doi.org/10.1016/j.im.2025.104107>.
- Hou, L., Pan, Y., & Zhu, J. J. (2021). Impact of scientific, economic, geopolitical, and cultural factors on international research collaboration. *Journal of Informetrics*, 15(3), Article 101194. <http://dx.doi.org/10.1016/j.joi.2021.101194>.
- Ji, S., Feng, Y., Ji, R., Zhao, X., Tang, W., & Gao, Y. (2020). Dual channel hypergraph collaborative filtering. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 2020–2029). Virtual Event, CA, USA and New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3394486.3403253>.
- Jiang, Y., Ma, H., Liu, Y., & Li, Z. (2020). Exploring user trust and reliability for recommendation: A hypergraph ranking approach. In H. Yang, K. Pasupa, A. C.-S. Leung, J. T. Kwok, J. H. Chan, & I. King (Eds.), *Neural information processing* (pp. 333–344). Cham: Springer International Publishing, http://dx.doi.org/10.1007/978-3-030-63833-7_28.
- Kim, E.-S., Kang, W. Y., On, K.-W., Heo, Y.-J., & Zhang, B.-T. (2020). Hypergraph attention networks for multimodal learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14581–14590).
- Konstas, I., Stathopoulos, V., & Jose, J. M. (2009). On social networks and collaborative recommendation. In *SIGIR '09, Proceedings of the 32nd international ACM SIGIR conference on research and development in information retrieval* (pp. 195–202). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/1571941.1571977>.
- Kouki, P., Schaffer, J., Pujara, J., O'Donovan, J., & Getoor, L. (2019). Personalized explanations for hybrid recommender systems. In *Proceedings of the 24th international conference on intelligent user interfaces* (pp. 379–390). Marina del Ray, California and New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3301275.3302306>.
- Lerner, J., & Hăncean, M.-G. (2023). Micro-level network dynamics of scientific collaboration and impact: Relational hyperevent models for the analysis of coauthor networks. *Network Science*, 11(1), 5–35. <http://dx.doi.org/10.1017/nws.2022.29>.
- Li, X., Guo, R., Chen, J., Hu, Y., Qu, M., & Jiang, B. (2023). Effective hybrid graph and hypergraph convolution network for collaborative filtering. *Neural Computing and Applications*, 35(3), 2633–2646. <http://dx.doi.org/10.1007/s00521-022-07735-y>.
- Li, K., Zhang, Y., Li, X., Yuan, M., & Zhou, W. (2025). Mask diffusion-based contrastive learning for knowledge-aware recommendation. *IEEE Transactions on Knowledge and Data Engineering*, 37(9), 5407–5419. <http://dx.doi.org/10.1109/TKDE.2025.3582767>.
- Liang, D., Krishnan, R. G., Hoffman, M. D., & Jebara, T. (2018). Variational autoencoders for collaborative filtering. In *WWW '18, Proceedings of the 2018 world wide web conference* (pp. 689–698). Republic and Canton of Geneva, CHE: International World Wide Web Conferences Steering Committee, <http://dx.doi.org/10.1145/3178876.3186150>.
- Lin, J., Dai, X., Xi, Y., Liu, W., Chen, B., Zhang, H., Liu, Y., Wu, C., Li, X., Zhu, C., Guo, H., Yu, Y., Tang, R., & Zhang, W. (2025). How can recommender systems benefit from large language models: A survey. *ACM Transactions on Information Systems*, 43(2), <http://dx.doi.org/10.1145/3678004>.
- Liu, J., Guo, X., Xu, S., Bu, Y., Sugimoto, C. R., Larivière, V., Song, Y., & Zhou, H. (2024). Understanding super-partnerships in scientific collaboration: Evidence from the field of economics. *Journal of the Association for Information Science and Technology*, 75(6), 717–733. <http://dx.doi.org/10.1002/asi.24876>.
- Liu, X., Yin, D., Zheng, J., Zhang, X., Zhang, P., Yang, H., Dong, Y., & Tang, J. (2022). OAG-BERT: Towards a unified backbone language model for academic knowledge services. In *KDD '22, Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 3418–3428). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3534678.3539210>.
- Ma, H., Zhang, D., Zhao, W., Wang, Y., & Shi, Z. (2018). Leveraging hypergraph random walk tag expansion and user social relation for microblog recommendation. In *2018 IEEE international conference on data mining* (pp. 1158–1163). <http://dx.doi.org/10.1109/ICDM.2018.00152>.
- Peng, X., & Shi, J. (2024). The drivers, features, and influence of first scientific collaboration among core scholars from Chinese library and information field. *Journal of the Association for Information Science and Technology*, 75(11), 1268–1287. <http://dx.doi.org/10.1002/asi.24888>.
- Porter, A. L., Cohen, A. S., David Roessner, J., & Perreault, M. (2007). Measuring researcher interdisciplinarity. *Scientometrics*, 72(1), 117–147. <http://dx.doi.org/10.1007/s11192-007-1700-5>.
- Shen, Y., Wu, Y., Zhang, Y., Shan, C., Zhang, J., Letaief, B. K., & Li, D. (2021). How powerful is graph convolution for recommendation? In *CIKM '21, Proceedings of the 30th ACM international conference on information & knowledge management* (pp. 1619–1629). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3459637.3482264>.
- Silva Rodrigues, P. H., de Sousa, D. X., França, C., Rabbi, G., Couto Rosa, T., & Gonçalves, M. A. (2025). Risk-sensitive optimization of neural deep learning ranking models with applications in Ad-Hoc retrieval and recommender systems. *Information Processing & Management*, 62(4), Article 104126. <http://dx.doi.org/10.1016/j.ipm.2025.104126>.
- Tang, J., Zhang, J., Yao, L., Li, J., Zhang, L., & Su, Z. (2008). ArnetMiner: Extraction and mining of academic social networks. In *KDD '08, Proceedings of the 14th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 990–998). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/1401890.1402008>.
- Tortoriello, M., McEvily, B., & Krackhardt, D. (2015). Being a catalyst of innovation: The role of knowledge diversity and network closure. *Organization Science*, 26(2), 423–438. <http://dx.doi.org/10.1287/orsc.2014.0942>.
- van der Maaten, L., & Hinton, G. (2008). Visualizing data using T-SNE. *Journal of Machine Learning Research*, 9(86), 2579–2605.
- Wang, X., He, X., Wang, M., Feng, F., & Chua, T.-S. (2019). Neural graph collaborative filtering. In *SIGIR '19, Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval* (pp. 165–174). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3331184.3331267>.
- Wang, J., Liang, Y., Meng, F., Sun, Z., Shi, H., Li, Z., Xu, J., Qu, J., & Zhou, J. (2023). Is ChatGPT a good NLG evaluator? A preliminary study. In Y. Dong, W. Xiao, L. Wang, F. Liu, & G. Carenini (Eds.), *Proceedings of the 4th new frontiers in summarization workshop* (pp. 1–11). Singapore: Association for Computational Linguistics, <http://dx.doi.org/10.18653/v1/2023.newsum-1.1>.
- Wang, W., Tang, T., Xia, F., Gong, Z., Chen, Z., & Liu, H. (2022). Collaborative filtering with network representation learning for citation recommendation. *IEEE Transactions on Big Data*, 8(5), 1233–1246. <http://dx.doi.org/10.1109/TBDATA.2020.3034976>.
- Wang, X., Zhang, R., Zhao, B., Yao, Y., Zhao, H., & Zhu, X. (2024). Medical knowledge graph completion via fusion of entity description and type information. *Artificial Intelligence in Medicine*, 151, Article 102848. <http://dx.doi.org/10.1016/j.artmed.2024.102848>.
- Wu, L., Yi, F., Bu, Y., Lu, W., & Huang, Y. (2024). Toward scientific collaboration: A cost-benefit perspective. *Research Policy*, 53(2), Article 104943. <http://dx.doi.org/10.1016/j.respol.2023.104943>.

- Xia, L., Huang, C., Xu, Y., Zhao, J., Yin, D., & Huang, J. (2022). Hypergraph contrastive collaborative filtering. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval* (pp. 70–79). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3477495.3532058>.
- Xie, Z., Ouyang, Z., Li, J., Dong, E., & Yi, D. (2018). Modelling transition phenomena of scientific coauthorship networks. *Journal of the Association for Information Science and Technology*, 69(2), 305–317. <http://dx.doi.org/10.1002/asi.23935>.
- Yadati, N., Nimishakavi, M., Yadav, P., Nitin, V., Louis, A., & Talukdar, P. (2019). HyperGCN: A new method for training graph convolutional networks on hypergraphs. vol. 32, In *Advances in neural information processing systems*. Curran Associates, Inc.
- Yu, B., Zhang, J., Fu, Y., & Xu, Z. (2025). A weighted heterogeneous graph attention network method for purchase prediction of potential consumers with multibehaviors. *Information Processing & Management*, 62(5), Article 104175. <http://dx.doi.org/10.1016/j.ipm.2025.104175>.
- Zhang, C., Bu, Y., Ding, Y., & Xu, J. (2018). Understanding scientific collaboration: Homophily, transitivity, and preferential attachment. *Journal of the Association for Information Science and Technology*, 69(1), 72–86. <http://dx.doi.org/10.1002/asi.23916>.
- Zhang, Y., & Chen, X. (2020). Explainable recommendation: A survey and new perspectives. *Foundations and Trends in Information Retrieval*, 14(1), 1–101. <http://dx.doi.org/10.1561/15000000066>.
- Zhang, F., Shi, S., Zhu, Y., Chen, B., Cen, Y., Yu, J., Chen, Y., Wang, L., Zhao, Q., Cheng, Y., Han, T., An, Y., Zhang, D., Tam, W. L., Cao, K., Pang, Y., Guan, X., Yuan, H., Song, J., Tang, J. (2024). OAG-bench: A human-curated benchmark for academic graph mining. In *KDD '24, Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 6214–6225). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3637528.3672354>.
- Zhang, Y., Wu, M., Zhang, G., & Lu, J. (2023). Stepping beyond your comfort zone: diffusion-based network analytics for knowledge trajectory recommendation. *Journal of the Association for Information Science and Technology*, 74(7), 775–790. <http://dx.doi.org/10.1002/asi.24754>.
- Zhang, Y., Zhang, Y., Liao, W., Li, X., & Wang, X. (2025). Multi-view self-supervised learning on heterogeneous graphs for recommendation. *Applied Soft Computing*, 174, Article 113056. <http://dx.doi.org/10.1016/j.asoc.2025.113056>.
- Zheng, L., Lu, C.-T., Jiang, F., Zhang, J., & Yu, P. S. (2018). Spectral collaborative filtering. In *recSys '18, Proceedings of the 12th ACM conference on recommender systems* (pp. 311–319). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3240323.3240343>.
- Zhou, D., Huang, J., & Schölkopf, B. (2006). Learning with hypergraphs: Clustering, classification, and embedding. vol. 19, In *Advances in neural information processing systems*. MIT Press.
- Zhu, Y., Guan, Z., Tan, S., Liu, H., Cai, D., & He, X. (2016). Heterogeneous hypergraph embedding for document recommendation. *Neurocomputing*, 216, 150–162. <http://dx.doi.org/10.1016/j.neucom.2016.07.030>.
- Zou, B., Wang, Y., Kwok, C. K., & Cen, Y. (2023). Directed collaboration patterns in funded teams: A perspective of knowledge flow. *Information Processing & Management*, 60(2), Article 103237. <http://dx.doi.org/10.1016/j.ipm.2022.103237>.